

一、练习 9.2.2

可用表达式分析

全集

$$E = \{ a+b, c-a, b+d, b*d, a-d \}$$

基本块的 e_gen / e_kill

B1

$$(1) a=1 \quad (2) b=2$$

$$e_gen = \emptyset$$

$$e_kill = \{ a+b, c-a, b+d, b*d, a-d \}$$

B2

$$(3) c=a+b \quad (4) d=c-a$$

$$e_gen = \{ a+b, c-a \}$$

$$e_kill = \{ b+d, b*d, a-d \}$$

B3

$$(5) d=b+d$$

$$e_gen = \{ b+d \}$$

$$e_kill = \{ c-a, a-d, b*d \}$$

B4

$$(6) d=a+b \quad (7) e=e+1$$

$$e_gen = \{ a+b \}$$

$$e_kill = \{ c-a, b+d, b*d, a-d \}$$

B5

$$(8) b=a+b \quad (9) e=c-a$$

$$e_gen = \{ a+b, c-a \}$$

$$e_kill = \{ b+d, b*d, a-d \}$$

B6

$$(10) a=bd \quad (11) b=a-d$$

$$e_gen = \{ bd, a-d \}$$

$$e_kill = \{ a+b, c-a, b+d \}$$

IN / OUT (迭代到不动点)

B1

IN = $\emptyset$

OUT = $\emptyset$

B2

IN = $\emptyset$

OUT = { a+b , c-a }

B3

IN = { a+b , c-a }

OUT = { b+d }

B4

IN = { b+d }

OUT = { a+b }

B5

IN = { a+b }

OUT = { a+b , c-a }

B6

IN = { a+b , c-a }

OUT = { b*d , a-d }

二、练习 9.2.3

活跃变量分析

def / use

B1

def = { a , b }

use = $\emptyset$

B2

def = { c , d }

$\text{use} = \{ a, b \}$

B3

$\text{def} = \{ d \}$

$\text{use} = \{ b, d \}$

B4

$\text{def} = \{ d, e \}$

$\text{use} = \{ a, b, e \}$

B5

$\text{def} = \{ b, e \}$

$\text{use} = \{ a, c \}$

B6

$\text{def} = \{ a, b \}$

$\text{use} = \{ d \}$

IN / OUT (反向, 不动点)

B6

$\text{OUT} = \emptyset$

$\text{IN} = \{ d \}$

B5

$\text{OUT} = \{ d \}$

$\text{IN} = \{ a, c, d \}$

B4

$\text{OUT} = \{ a, c, d \}$

$\text{IN} = \{ a, b, c, e \}$

B3

$\text{OUT} = \{ a, b, c, e \}$

$\text{IN} = \{ a, b, c, d, e \}$

B2

$\text{OUT} = \{ a, b, c, d, e \}$

$\text{IN} = \{ a, b, e \}$

B1

OUT = { a , b , e }

IN = { e }